Magpie — Sprint 12 spec: Command-centre dashboard (Phase 6, final sprint)

Audience: Codex (implementation). **Author:** Claude. **Status: ready to implement.** **Locks:** D-103…D-109. **References:** Q-17 (converged), OQ-3b (resolved — ChatGPT verdict 2026-06-15), D-90 / D-94 (dashboards = live DRF aggregates, no warehouse), D-91 / D-98 (sale provenance + never-fabricate-profit), F-09 (no warehousing of eBay Browse data). Design direction grounded via the `frontend-design` skill.

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1. Goal (one paragraph)

A single, beautiful, **useful** command-centre dashboard that tells Regan how the business is doing at a glance: realised profit, what's selling, how accurate his valuations are, and what to list next. The **top KPI row is personal** (the user picks the tiles); **everything below it is fixed and polished**. This is a dashboard, **not** a dashboard-construction toolkit — no drag-and-drop builder, no report designer. It must launch **honest at small sample** (Magpie has ~4 real sales once the staged eBay orders are triaged), so empty and small-sample states are first-class, not afterthoughts. All numbers computed **live from the DB** (DRF aggregates) — no analytics warehouse, no caching layer in v1.

2. Scope

IN (v1):

- One command-centre page (vertical scroll).
- A **configurable KPI tile row** (pick 4–5 from a fixed allow-list, reorderable, backend-persisted).
- Fixed dashboard sections below the KPI row (see §6).
- The **estimate-vs-actual signature visualisation** (§7).
- Global filters: time range, category, channel, include/exclude external-or-cost-basis-unknown (§5).
- Live DRF aggregate endpoints (§4).
- First-class empty / small-sample / loading states everywhere (§8).

- Deliberate mobile reflow (§9).

OUT (explicitly, v1):

- ❌ Drag-and-drop dashboard layout editor.
- ❌ Movable / configurable panels **below** the KPI row (those are fixed in v1).
- ❌ Report builder / custom-query UI.
- ❌ Drill-down into a metric's underlying rows (deferred to **v1.1**).
- ❌ Dark mode (light-first; dark later only if it's nearly free — don't spend the sprint on it).
- ❌ Any analytics warehouse / materialised aggregates / background jobs.

3. Data model (one additive migration)

`DashboardPreference` — a **singleton** app-setting record (single-user runtime per F-14; created with defaults on first read, never requires the user to "set up" anything):

- `kpi_tiles` — JSON, an **ordered** list of tile IDs, e.g. `["realised_profit","net_proceeds","sell_through","items_sold","avg_realised_margin"]`. Order is display order.
- `schema_version` — int (default 1), so future tile-catalogue changes can migrate safely.
- standard timestamps.
- **Validation on write:** every ID must be in the allow-list (§5.2); unknown IDs are **stripped** (don't 400 the whole save over one stale ID — strip and save the valid remainder); enforce **max 5**, **min 3** (if a save would drop below 3, reject with a clear message); de-duplicate.
- **Read with no record → return the default row** (§5.1), don't auto-create until the user actually saves a change.

No other model changes. `quantity_remaining`, valuations, fees etc. already exist.

4. Backend — live aggregate endpoints (DRF, read-only)

All under `/api/analytics/` unless noted. All accept the global filters (§5.3) as query params. All are **pure reads** computed from current data (`SaleRecord`, `ValuationReport` where `is_current`, `InventoryItem`, `EbayOrderStaging`). No writes except the preference endpoint. Keep queries set-based (annotate/aggregate), not per-row Python loops.

- `GET /api/analytics/summary` → every KPI tile value in the allow-list (the frontend renders only the selected subset, but compute all so the picker can preview any), plus the small action-counts. Cheap at this scale.
- `GET /api/analytics/pnl` → realised-profit and net-proceeds **time series by month** for the P&L section.
- `GET /api/analytics/by-category` → revenue / realised margin / sell-through **per category**.
- `GET /api/analytics/estimate-vs-actual` → the scatter payload (per sold item: estimated value at sale time vs actual), the **accuracy metric**, and the **estimated-vs-actual fees** comparison (§7).
- `GET /api/analytics/aging` → available-inventory **age buckets** (0–30 / 31–90 / 91–180 / 180+ days) with counts and summed cost basis / estimated value.
- `GET /api/analytics/listing-opportunities` → **what's worth listing**: available items **not currently listed/drafted**, ranked by current estimated value (and estimated margin where cost basis is known).
- `GET|PUT /api/dashboard/preferences` → the `DashboardPreference` (GET returns defaults if no record; PUT validates per §3).

5. The configurable KPI row (OQ-3b)

5.1 Default row (ships out of the box, used when no preference record)

`realised_profit`, `net_proceeds`, `sell_through`, `items_sold`, `avg_realised_margin`.

5.2 Allow-list (the candidate tile pool) + definitions

Compute over the **filtered** period. **Honest-profit rule (D-98): never fabricate profit.** Sales flagged `cost_basis_unknown` (incl. `is_external` with unknown cost) contribute to **revenue/fees** but are **excluded from profit/margin**; show the excluded count as a secondary line where relevant.

Tile ID	Label (end-user)	Definition / source
`realised_profit`	Realised profit	Σ `SaleRecord.realised_profit`, **known-cost sales only**. Secondary: "+N sales with unknown cost

		excluded."
`gross_revenue`	Revenue	Σ sale price × qty. **Includes** external/unknown-cost (revenue is known).
`net_proceeds`	Net proceeds	Σ `net_proceeds` (price – fees – shipping). Includes external (fees known).
`items_sold`	Items sold	Σ quantity sold (count of sale lines × qty).
`sell_through`	Sell-through	sold ÷ (sold + currently-available), as a %. *(Reasonable v1 proxy — refinable.)*
`avg_realised_margin`	Avg margin	**aggregate** margin = Σ realised_profit ÷ Σ revenue over **known-cost** sales (more stable than averaging per-sale ratios).
`avg_time_to_sale`	Avg time to sale	mean (sale_date – item acquisition/created date) in days, sold items with a known acquisition date.
`inventory_cost_basis`	Inventory cost	Σ cost basis of currently-available items (× remaining qty).
`estimated_inventory_value`	Est. inventory value	Σ current valuation × remaining qty, available items.
`aged_inventory_count`	Aged stock	count of available items older than **90 days** (v1 threshold).
`unresolved_ebay_staging_count`	eBay to triage	count of pending `EbayOrderStaging` rows.
`cost_basis_unknown_sales_count`	Unknown-cost sales	count of sales where profit isn't computable.

5.3 Global filters

- **Time range** — presets (This month / Last 3 / 6 / 12 months / All) + custom range.

- **Category** — multi-select.
- **Channel** — eBay / manual / external / all.
- **Include/exclude external-or-cost-basis-unknown** — default = **include in revenue & fees, exclude from profit** (matches D-98); the toggle lets Regan fully hide them or fully include for inspection.

5.4 Picker behaviour

- Choose **4–5** tiles (hard max 5, min 3); default selection = §5.1.
- **Reorder** via simple up/down controls (ordering is just the array order — cheap; no drag-and-drop library needed).
- Persisted **server-side** (§3) so desktop and phone always match. No localStorage as the source of truth.
- Picker is a small "Customise" affordance on the dashboard (e.g. a popover/sheet), not a separate page.

6. Fixed sections below the KPI row (deliberate order)

1. **Realised P&L over time** — line/area of realised profit + net proceeds by month. Profit series excludes unknown-cost sales (note if any excluded).
2. **Estimate vs Actual (the signature, §7).**
3. **Revenue / net proceeds** — bars by month (or by channel).
4. **Margin by category** — bars, known-cost sales.
5. **Sell-through & time-to-sale** — compact stat cards (optionally per category).
6. **Aging inventory** — age buckets (§4 `aging`), aged stock highlighted.
7. **What's worth listing** — ranked table (§4 `listing-opportunities`), action-oriented: the next best items to list.

7. Signature visualisation — "How good are my valuations?"

A **scatter plot**: each point = a sold item where a valuation snapshot existed at sale time (per D-91). **x = estimated value, y = actual sale price**; a **45° reference line** = perfect prediction. Above the line = sold for more than estimated; below = less.

- **Paired with:** an **accuracy KPI** (e.g. % of sales within ±20% of estimate, or median absolute % error — pick one and label it plainly) and an **estimated-vs-actual fees**

mini-comparison (`estimated_fee_snapshot` vs `actual_fees_total`).

- **Data-aware (critical — launches at N≈4):** with few points, render the points + line but annotate **"small sample (N=4)"** and do **not** draw trend lines or imply significance. At N=0, show the designed empty state (§8), not an empty chart frame.
- **Phone:** a scatter is unreadable on a narrow screen — **fall back to a compact table** (Item · Estimated · Actual · Δ%) below a small accuracy summary.

8. Empty / small-sample / loading states (first-class)

Every section ships three states. Write copy as **direction, not apology** (frontend-design skill): explain what's missing and the action that fills it.

- **Empty (N=0):** e.g. P&L — *"No sales yet. Resolve your eBay orders or log a sale and your profit lands here."* (link to `/ebay/orders` or `/sales`). Listing opportunities — *"Nothing waiting to be listed. Add inventory or run a valuation to see what's worth listing."*
- **Small sample:** show the data **plus** a quiet "small sample" marker; never fabricate trends.
- **Loading:** skeletons, not spinners-on-blank; no layout shift when data arrives. The dashboard must look **intentional and calm at N=4**, not broken or sparse.

9. Design direction (light-first dealer's-ledger; refine via the `frontend-design` skill)

Run the skill's two-pass plan → self-critique before building. **Avoid the AI-default looks** (cream + high-contrast serif + terracotta; near-black + neon accent; broadsheet hairlines-everywhere). The identity below is grounded in the subject — a precious-metals & collectibles **dealer's ledger**: paper, ink, hairline rules, **tabular figures**, one restrained **antique-gold** accent (gold is in the business name), profit/loss read in muted ledger greens/reds. Spend the boldness on the **signature scatter**; keep everything else quiet.

Starting tokens (Codex may refine, Regan signs off):

- **Palette (light-first):** cool paper `#FAFAF7` (not warm cream), ink `#1C1B19` , antique gold accent `#9A7B2E` (used sparingly — active states, the 45° line, key emphasis), slate ink-secondary `#2C3340` , hairline `#E4E1D8` ; semantic **profit** `#2E7D5B` / **loss** `#B0413E` (muted, not neon).
- **Type roles:** a **characterful display** face for the page title + big KPI numbers (used with restraint); a clean **body** sans; and a **tabular/mono** face for all figures so

columns align like a ledger (this mono-for-data choice is the subject-appropriate move, not decoration). Set a real type scale. Don't reach for a generic high-contrast serif.

- **Layout:** title + global filters at top → the configurable KPI row → fixed sections; the scatter gets visual prominence. Calm spacing, hairline separators, data-dense but unhurried.
- **Signature:** the estimate-vs-actual scatter + gold 45° line is the one memorable element.
- **Quality floor (non-negotiable):** responsive to phone, visible keyboard focus, `prefers-reduced-motion` respected, no layout shift on load.

10. Acceptance criteria (closure gate)

Backend / data:

- Aggregate-endpoint tests with realistic fixtures: multi-category, multi-month, **external + cost-basis-unknown** sales, and a **small-sample (N≈4)** case. Assert the **honest-profit rule** (external/unknown in revenue & fees, **never** in profit/margin) and correct filter behaviour.
- `preferences` : GET returns the **default row when no record**; PUT **validates against the allow-list** (unknown IDs stripped, max 5 / min 3 enforced, de-duped) and **persists**.

KPI row (OQ-3b additions):

- Default KPI row appears with **no preference record**.
- Changing selected tiles **persists**; **reloading keeps** them; **phone and desktop use the same saved set**.
- **Invalid/unknown tile IDs are ignored or reset safely** (no crash, no blank row).
- Reorder works (stored order = display order).

States & mobile:

- Every section has working **empty / small-sample / loading** states; nothing looks broken at **N=0 or N=4**.
- Deliberate **mobile reflow**; the scatter **falls back to a table** on phone; quality floor met.

Explicitly verify NOT present: drag-and-drop layout editor; movable panels below the

KPI row; report-builder.

Process: `npm run test` + `typecheck` + `build` green; backend tests green; **dual-lane `Validation` (SQLite + Postgres) green**; rebuild + restart the `Magpie` service so the live app serves it; **desktop AND phone screenshots** captured; **Regan rubric sign-off** (§11) closes the sprint.

11. Regan's sign-off rubric (the aesthetic gate)

- Does it read as a **serious, calm command-centre** — and clearly **not** a generic template or the AI-default dark-neon / cream-serif look?
- Is the **gold accent restrained** and the **data legible** (figures align, tabular)?
- Is the **estimate-vs-actual scatter** the obvious signature — prominent and readable?
- Do the **empty / small-sample states** feel intentional and **invite action** (not broken, not apologetic)?
- Does it **work on the phone** (tiles reflow, scatter→table, sections stack)?
- Are the **numbers honest** — no fabricated profit, external/unknown handled visibly?

12. For Codex

Reasoning: High. Additive migration only (`DashboardPreference`); **no destructive migration**, no warehouse, no background jobs, analytics endpoints are **read-only** (only the preference PUT writes). Don't print or commit secrets, `.env` , eBay secrets, OAuth codes/tokens, rclone config, backup passphrase, or encryption keys. Commit evidence (tests + desktop/phone screenshots) as `SPRINT12_EVIDENCE.md` . This is the **last Phase-6 sprint** — closing it completes Phase 6.